

Per-Tetrahedron vs. Nodal-Averaged Volume Penalty for Tetrahedral Mass-Spring Models

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1 Introduction

Tetrahedral mass-spring models (MSMs) approximate linear-elastic solids by replacing continuum elements with networks of harmonic springs on tetrahedral meshes. A well-known limitation of central-force spring networks is that the Cauchy relations constrain the Poisson’s ratio to $\nu = 1/4$ in three dimensions [1, 4]. To achieve arbitrary ν , a volumetric penalty term is added to each tetrahedron [2, 3].

For finite element methods on tetrahedral meshes, it is known that per-element volume constraints can cause *volumetric locking*—artificial stiffening as $\nu \rightarrow 0.5$ —because the number of element constraints exceeds the nodal degrees of freedom. Bonet and Burton [5] proposed an average nodal pressure (ANP) formulation to alleviate this. This report compares two implementations of the volume penalty within the custom `improper_style tetmsm` in LAMMPS: a per-tetrahedron formulation and a nodal-averaged formulation, validated through uniaxial tension of a beam.

2 Model

2.1 Spring energy

Each edge (i, j) of the tetrahedral mesh carries a bond energy

$$U_{\text{bond}} = \frac{1}{2} \kappa_E r_0 \left(1 - \frac{r}{r_0}\right)^2, \quad (1)$$

where r_0 is the reference edge length, r is the current length, and κ_E (energy/length) is determined from Lloyd’s least-squares identification [1]. For a tetrahedron Δ with equivalent edge length $\tilde{l}_\Delta = (12V_\Delta/\sqrt{2})^{1/3}$, the per-tet spring stiffness contribution is $\kappa_E = (\sqrt{2}/5) G \tilde{l}_\Delta$, where G is the shear modulus. With springs alone, $\nu = 1/4$ identically.

2.2 Per-tetrahedron volume penalty

For each tetrahedron Δ with signed volume

$$V_\Delta = \frac{1}{6} \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}), \quad (2)$$

the per-tet volume penalty energy is

$$U_\Delta^{\text{vol}} = \frac{1}{2} \kappa_V V_{0,\Delta} \left(1 - \frac{V_\Delta}{V_{0,\Delta}}\right)^2, \quad (3)$$

where $V_{0,\Delta}$ is the reference volume and the volume stiffness is

$$\kappa_V = \frac{G(4\nu - 1)}{1 - 2\nu}. \quad (4)$$

The force on vertex k of tetrahedron Δ is

$$\mathbf{f}_k = -\kappa_V \frac{V_\Delta - V_{0,\Delta}}{V_{0,\Delta}} \frac{\partial V_\Delta}{\partial \mathbf{x}_k}, \quad (5)$$

where $\partial V_\Delta / \partial \mathbf{x}_k$ are the standard face-area vectors (cross products of edge vectors divided by 6). At $\nu = 1/4$, $\kappa_V = 0$ and the model reduces to Lloyd's springs-only MSM.

2.3 Nodal-averaged volume penalty

Following the ANP concept [5], define a nodal volume for each atom i :

$$V_i = \frac{1}{4} \sum_{\Delta \in \mathcal{T}_i} V_\Delta, \quad (6)$$

where $\mathcal{T}_i$ is the set of tetrahedra sharing node i . The nodal volumetric strain is $\varepsilon_i = (V_i - V_{0,i})/V_{0,i}$, and the energy per node is

$$U_i^{\text{vol}} = \frac{1}{2} \kappa_V V_{0,i} \varepsilon_i^2. \quad (7)$$

Differentiating the total energy $U = \sum_j U_j$ yields the force on node k of tetrahedron Δ as

$$\mathbf{f}_k^{(\Delta)} = -\frac{S_\Delta}{24} \frac{\partial V_\Delta}{\partial \mathbf{x}_k}, \quad S_\Delta = \kappa_V \sum_{v \in \text{verts}(\Delta)} \varepsilon_v. \quad (8)$$

The nodal scheme replaces each tet's individual volumetric strain with the sum of its four vertex strains, coupling the constraint across neighboring elements. The implementation requires two passes over the improper list: one to accumulate nodal volumes (with reverse/forward communication for ghost atoms) and one to compute forces.

3 Validation: uniaxial beam tension

3.1 Setup

A rectangular beam of dimensions $0.4 \times 0.4 \times 2.4$ m (aspect ratio 1:1:6) is meshed with structured tetrahedra at resolution $N = 8$ elements along the shortest side. The material has shear modulus $G = 4000$ Pa, density $\rho = 1000$ kg/m³ (irrelevant for quasi-static loading), and the Young's modulus follows from $E = 2G(1 + \nu)$. Poisson's ratio ν is swept as the independent variable.

Boundary conditions: the bottom face ($z = 0$) is fully clamped (`setforce 0 0 0`), and the top face ($z = 2.4$ m) is displaced incrementally in z while remaining free in x and y (`setforce NULL NULL 0`). Lateral faces are traction-free. The beam is loaded quasi-statically in 100 increments to a maximum axial strain of $\varepsilon_z^{\text{max}} = 5\%$, with energy minimization at each step.

3.2 Measurement

The Young's modulus is extracted globally from

$$E_{\text{meas}} = \frac{\sigma_z}{\varepsilon_z}, \quad \sigma_z = \frac{F_z^{\text{bot}}}{A_0}, \quad (9)$$

where F_z^{bot} is the total reaction force on the clamped face and A_0 is the initial cross-sectional area measured from the mid-section. The axial strain $\varepsilon_z = (L - L_0)/L_0$ uses the full beam length.

Poisson's ratio is measured in the mid-section ($0.8 \leq z \leq 1.6$ m) to avoid Saint-Venant end effects. A least-squares regression of lateral displacement u_x against reference coordinate x_0 yields the lateral strain ε_x :

$$\varepsilon_x = \frac{\text{Cov}(u_x, x_0)}{\text{Var}(x_0)}, \quad \nu_{\text{meas}} = -\frac{\varepsilon_x}{\varepsilon_z}, \quad (10)$$

and analogously for ε_y . The reported G_{meas} is derived from E_{meas} and ν_{meas} via $G = E/[2(1 + \nu)]$.

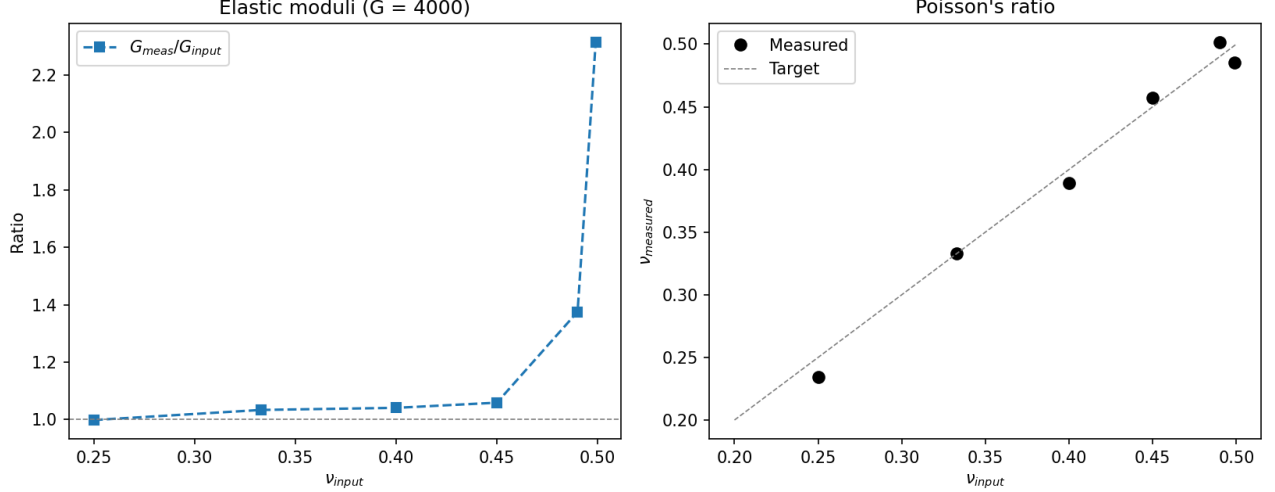


Figure 1: Per-tetrahedron volume penalty. Left: ratio $G_{\text{meas}}/G_{\text{input}}$ vs. ν_{input} . Right: measured vs. target Poisson’s ratio. The model accurately reproduces both G and ν for $\nu \lesssim 0.45$. Volumetric locking is apparent near $\nu = 0.5$, where G_{meas} rises sharply.

4 Results and discussion

Figure 1 shows results for the per-tetrahedron formulation. The shear modulus ratio $G_{\text{meas}}/G_{\text{input}}$ remains close to unity for $\nu \leq 0.45$, confirming that the springs correctly reproduce G and that the volume penalty does not corrupt the shear response in this range. The measured Poisson’s ratio tracks the target line with only a slight under-prediction, consistent with the finite mesh resolution ($N = 8$). However, as $\nu \rightarrow 0.5$, volumetric locking sets in: G_{meas} rises to $\sim 2.3 G_{\text{input}}$ at $\nu = 0.5$, and ν_{meas} saturates near 0.5. This is expected, since the number of per-tet volume constraints (approximately 5–6 per node for structured meshes) exceeds the three translational degrees of freedom per node.

Figure 2 shows results for the nodal-averaged formulation. The behavior is qualitatively different: $G_{\text{meas}}/G_{\text{input}}$ decreases monotonically from ~ 1.13 at $\nu = 0$ to ~ 0.78 at $\nu = 0.5$, crossing unity near $\nu = 0.25$. The measured Poisson’s ratio is severely under-predicted at all ν , dropping below 0.05 at $\nu = 0.5$ —the opposite of the expected near-incompressible response.

The poor performance of the nodal formulation can be understood from the energy structure. In the per-tet version, the total volumetric energy is $\sum_{\Delta} \frac{1}{2} \kappa_V V_{0,\Delta} (\delta V_{\Delta}/V_{0,\Delta})^2$, which penalizes each tet’s volume change independently. In the nodal version, the energy $\sum_i \frac{1}{2} \kappa_V V_{0,i} \varepsilon_i^2$ averages volumes before squaring. By Jensen’s inequality, averaging before squaring is strictly softer than squaring then summing: the nodal penalty under-penalizes non-uniform volumetric deformation patterns. On a structured mesh where local volume fluctuations are systematic rather than random, this softening is severe enough to render the volume constraint ineffective.

In the finite element context, the ANP technique works because it replaces per-element *pressure* with a nodal average within a variationally consistent weak form. Transplanting only the averaging idea to an energy-based MSM penalty—without FEM’s variational consistency—produces a formulation that is too compliant. Calibrating κ_V to compensate would defeat the simplicity of the approach.

5 Conclusion

The per-tetrahedron volume penalty accurately reproduces both G and ν for $\nu \leq 0.45$ on a structured mesh with $N = 8$. Volumetric locking appears near $\nu = 0.5$, as expected. The nodal-averaged formulation produces systematically incorrect elastic moduli and Poisson’s ratio across the entire

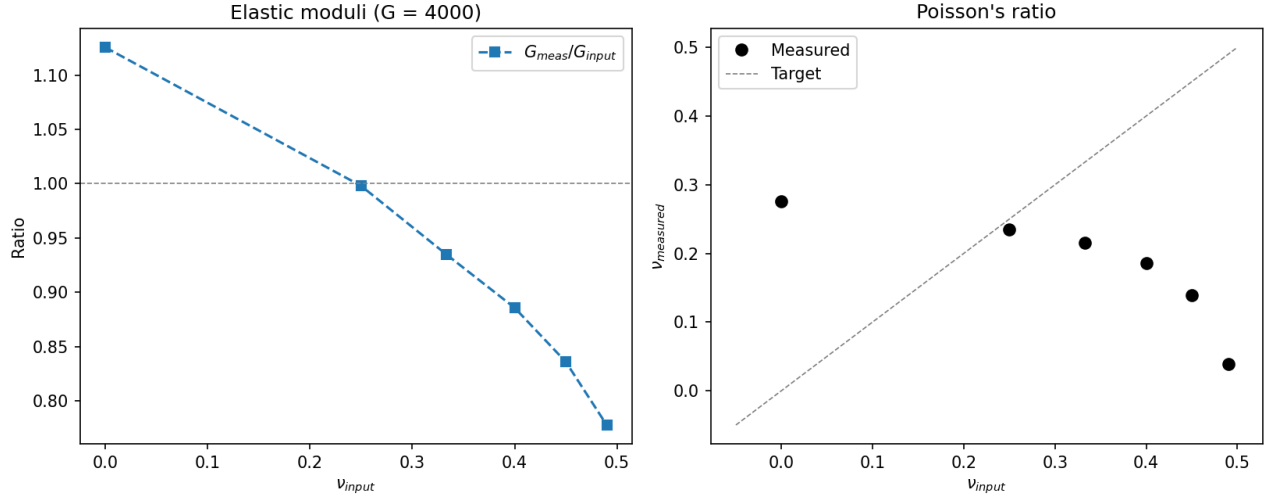


Figure 2: Nodal-averaged volume penalty. Left: $G_{\text{meas}}/G_{\text{input}}$ vs. ν_{input} . Right: measured vs. target Poisson's ratio. The shear modulus is systematically incorrect across the full range of ν , and the measured Poisson's ratio deviates strongly from the target.

range tested, and is therefore not adopted. Addressing near-incompressible behavior remains an open problem for MSMs.

References

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